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Project Title:	The Role of Inhibin and the Hypogonadal Axis in Regulating Bone Mass During Simulated Microgravity		
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Contact Monitor:		**Contact Phone:**	
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Task Description:

Bone loss during spaceflight is a serious obstacle to long-term space exploration and habitation. Studies demonstrate that microgravity decreases circulating testosterone. Early studies documented similar results in hindlimb suspended (HS) rats. However, these studies were complicated by the occurrence of cryptorchidism, which causes testicular atrophy and dysfunction. This inappropriate phenomenon exacerbates HS-related changes in the hypogonadal axis and may inappropriately contribute to bone loss during HS. Hence, we propose to examine the relative contributions of disuse and cryptorchidism to HS-induced bone loss and to test the efficacy of Inhibin, a member of the hypogonadal axis, as a skeletal countermeasure Task Objective: Determine whether cryptorchidism contributes to bone loss hindlimb suspended mice, a model of microgravity, and determine whether overexpression on Inhibin A prevents bone loss during hindlimb suspension in mice. Animals models should mimic the human situation as closely as possible. Cryptorchidism cannot occur in normal adult men. Hence, contribution to bone loss by cryptrochidism in HS mice would confound any data obtained using this model and requir refinement of the model to prevent cryptorchidism.
Countermeasures are still needed to prevent bone loss during microgravity. We have shown that Inhibin A can prevent bone loss following orchidectomy in mice. If Inhibin A can prevent bone loss during HS in mice, its signalling pathway will provide new targets for bone loss countermeasures during spaceflight.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits: Agents that can prevent bone loss during HS (simulated microgravity) have application to patients subjected to long-term bedrest, casting, transient paralysis, or other types of disuse that induce bone loss.

Task Progress:

All of the animal studies for this project are completed and samples are currently being analyzed.
AIM 1: In the 5 months since the current funding period began we have made significant progress on this aim. In summary, we have manufactured additional mouse-specific HS cages, completed a 3 week animal study, performed analyses of testicular and seminal vesicle atrophy, and evaluated the skeletal effect of HS by ex vivo bone marrow culture, microCT, and DEXA. The status of our ongoing analyses is discussed in detail below.

Experimental results:

In order to prevent cryptorchidism and the resulting endocrine deficits, such as steroid loss, during HS in mice, we initially used inguinal canal ligation (ICL) as described for the rat. During pilot studies it became clear that ICL was not feasible in the mouse, due to critical anatomical differences. Prior to thes studies, ICL for the prevention of cryptorchidism in the HS mouse had not been reported. Consequently, we utilized modified orchiopexy (OPX) to secure the testes within the scrotum.

For Aim 1, 16 adult (3.5 month old) Swiss-Webster mice underwent orchiopexy (OPX) to prevent cryptorchidism, while another 16 underwent sham surgery. OPX was performed using a single stitch of 6-0 prolene suture inserted through the distal end of the scrotum and both testis. For sham surgery, a single stitch was tied through only the scrotum. Eight mice in each group were hindlimb suspended (HS-Sham and HS-OPX), while the other eight were individually housed in standard cages (Con-Sham and Con-OPX). All mice were sacrificed after 3 weeks of HS, at which time the tails of all mice were pink, functional, and healthy over their entire length, demonstrating successful HS, with no obvious complications. Furthermore, no significant changes in body weight between any of the treatment groups were demonstrated.

Interestingly, HS induced only a modest 20% change in testicular weight in the mouse, compared with a 70% decrease in the HS rat. In addition, OPX (as performed here) did not have any effect on testicular or seminal vesicle weight.

Upon gross examination of the testes following 3 weeks of HS, it was obvious that the suture had pulled out of the testes during the experiment, thereby negating the effect of OPX. This prevents us from drawing any conclusions concerning our original hypothesis that the prevention of cryptorchidism will attenuate but not completely prevent HS-induced bone loss. However, important data have been obtained.

The striking species difference in the extent of testicular atrophy between rats and mice imply that the impact of cryptorchidism may be a concern only in the rat HS model. Completion of the endocrine and reproductive analyses in the mice is ongoing and required to fully validate this conclusion. Serum testosterone (T) assays, along with histology and Western blots of Inhibin B in testicular extracts will be performed by the end of Year 1, and will determine the degree of HS-induced testicular dysfunction.

In addition to the reproductive, several musculoskeletal analyses have been initiated and will be completed within year 1. DEXA was used to track changes in 2-dimensional bone mineral density (BMD) over the 3-week protocol and demonstrated the expected HS-induced decrease in BMD of the proximal tibia, distal femur, femoral midshaft, and total body. These findings are supported by detailed microCT analysis of trabecular bone volume and architecture in the proximal tibia where HS also induced significant bone loss and architectural deterioration.

In addition to the proposed analyses of the skeleton, reproductive tissues, and serum, we initiated analyses of hindlimb muscles which are also adversely affected by HS. The soleus and gastrocnemius muscles were harvested at sacrifice and atrophy was assessed by measuring muscle weight-to-body weight ratio. As with the other measures, analysis demonstrated that HS induced the expected atrophy in both the soleus and gastrocnemius (data not shown). In the next funding cycle we will continue to examine the effects of HS on soleus muscle gene expression (Aim 3).

While OPX in our hands was not successful, we have developed the HS mouse model in our laboratory and demonstrated the expected effects of HS on the skeleton and hindlimb muscles. Moreover, our data demonstrate that seminal vesicle weight is not significantly decreased by HS in the mouse, suggestin that testicular atrophy induced by HS in mice may not have a significant effect on systemic reproductive function. Further analysis is required to confirm this conclusion.

Progress in AIM 2

We have demonstrated that Inhibin A, an integral member of the hypogonadal hormone axis, increases bone mass in intact mice and prevents orchidectomy induced bone loss. These studies provide exciting evidence that Inhibin promotes bone anabolism in the presence or absence of androgens. Hence, we hypothesize that Inhibin overexpression will prevent the bone loss and increased fragility due to HS.

To date the animal study has been completed and analyses are currently in progress. Initial analysis of trabecular bone in the proximal using microCT has revealed that HS did produce the expected decrease in trabecular bone volume and deterioriation of architectural properties. In contrast, overexpression of Inhibin in HS mice reduced the degree of bone loss and prevented loss of trabecular number and increase in separation that occured in HS-Veh mice. This data demonstrates for the first time that InhA is capable of reducing bone loss during HS and suggests that the Inhibin signalling pathway may be a future traget for drug development.

Bibliography Type: Description: (Last Updated: 04/28/2011)

Abstracts for Journals and Proceedings

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